

Secondary Question Analysis

Q1. Which age group is most affected by air pollution-related health outcomes, and how does this vary by city?

Children and the elderly consistently come up as the two groups most affected across the available research. For children, the concern is mainly respiratory — a peer-reviewed study found that every 10-unit rise in AQI is associated with roughly a 1–1.4% increase in pediatric asthma-related emergency visits and hospital admissions. For the elderly, the larger risk is cardiovascular rather than respiratory, which matches how most global air pollution health literature separates the two groups.

City-level variation is harder to pin down precisely from public sources, but Delhi comes up repeatedly as the most severe case for child health impact, with several reports pointing to measurable lung function impairment in children living there long-term. Bengaluru has also reported a rise in asthma cases tied to air quality, though mostly from local news coverage rather than a peer-reviewed study.

Two figures from the original source list, 170K children died and 86% child/100-120 under age five, could not be verified against a clear, citable source, so they are left out rather than repeated as fact.

Sources

10-unit AQI increase and pediatric asthma ED visits: <https://pubmed.ncbi.nlm.nih.gov/35394903/>

Supporting study: <https://pubmed.ncbi.nlm.nih.gov/37628320/>

Supporting study: <https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2021.798746/full>

Supporting study: <https://www.mdpi.com/2077-0383/14/13/4659>

Air pollution in Delhi, child lung impairment: https://en.wikipedia.org/wiki/Air_pollution_in_Delhi

Bengaluru asthma cases and air quality: <https://timesofindia.indiatimes.com/city/bengaluru/asthma-cases-surge-thanks-to-poor-air-quality-late-diagnosis/articleshow/120905786.cms>

Q2. Who are the major competitors in the Indian air purifier market, and what are their key differentiators?

The market spans a wide price range. On the budget end, roughly ₹6,000–12,000, sit Eureka Forbes, Honeywell, and Xiaomi/Mi — compact units for single rooms, with HEPA filtration as the baseline and app connectivity increasingly common even here. Mid-range, ₹13,000–20,000, includes Kent, Qubo, and some Honeywell and Blueair models, generally adding a real-time AQI display and a dedicated activated-carbon stage for gases. At the premium end, ₹25,000–45,000 and up, Dyson and Coway lead, differentiated mainly by build quality, noise levels, and brand trust rather than filtration technology alone.

Annual filter replacement cost is a differentiator that isn't always obvious upfront — budget brands run ₹1,500–2,500 a year, premium brands can run ₹4,000–6,000 or more. A recurring complaint across reviews is that several models marketed as smart don't actually include a real PM sensor or an on-device display, using the word more as a label than a functioning feature.

Sources

Competitor and pricing reference sheet:

<https://docs.google.com/spreadsheets/d/1DdgyCVgwcnCgyq2XpvObXyybBrIXeAsG/edit>

Q3. What is the relationship between a city's population size and its average AQI, and do larger cities always suffer from worse air quality?

This question is answered directly from the project's own AQI and population datasets rather than external sources, shown as a scatter plot on the dashboard (population on one axis, average AQI on the other, one point per state, 2024 scope as specified in the brief).

The short answer is no, not always. A few states combine large population with genuinely high AQI, but several smaller-population states also show high AQI, and a few large states sit at a relatively low average AQI. Population size on its own is not a reliable predictor — it is one contributing factor among several, not a rule.

Sources

City population reference, used for metro-level market potential figures: <https://worldpopulationreview.com/cities/india>

Q4. How aware are Indian citizens of what AQI means, and do they understand its health implications?

The pattern across independent surveys is fairly consistent: most people have heard the term AQI, but fewer understand what it actually measures or what the categories mean. A 17-city perception study found awareness was highest in Tier 1 and metro cities, led by Delhi, but only around 35% of people who had heard of AQI could explain it correctly — and understanding of the specific terms AQI, PM2.5, and PM10 dropped to roughly 54%, 30%, and 18% respectively.

A separate Delhi-focused survey found an even sharper gap, with 93% of respondents unable to correctly explain the six AQI categories. Awareness also splits along income lines — one study found only about 10% of lower-income urban respondents were aware of the term, against 71% of middle-class respondents.

Sources

Survey response data:

<https://docs.google.com/spreadsheets/d/1Mo-UOH0Yag4urDcWfNHwHdm8um2Tpl6oaq30x8QQ1vU/edit>

Q5. Which pollution control policies introduced by the Indian government in the past five years have had the most measurable impact on improving air quality, and how have these impacts varied across regions or cities?

The National Clean Air Programme (NCAP) is the main policy framework here, tracked through CPCB's PRANA portal. Independent reporting shows a mixed picture rather than a clean success story — some coverage points to roughly a 27% drop in air pollution across Indian cities since 2019 on average, but the same coverage flags that the most polluted cities, including Delhi and Noida, have seen comparatively little improvement despite receiving NCAP funding, with fund utilisation in those cities reportedly staying below 40%.

In short, the policy exists and shows a measurable effect at the national average level, but its impact is uneven, and the cities that need it most have not shown the strongest gains so far.

Sources

CPCB PRANA portal, plan and results: <https://prana.cpcb.gov.in/#/about>

CPCB AQI bulletin: https://cpcb.nic.in//upload/Downloads/AQI_Bulletin_20250703.pdf

27% drop in air pollution since 2019, reporting: <https://www.downtoearth.org.in/pollution/indian-cities-see-27-drop-in-air-pollution-since-2019-but-usual-suspects-fare-poorly-report>

Fund utilisation limitations in worst-affected cities: <https://www.downtoearth.org.in/pollution/most-polluted-delhi-noida-among-cities-with-less-than-40-national-clean-air-programme-fund-utilisation-cse-assessment>

Live CPCB AQI data: https://airquality.cpcb.gov.in/AQI_India/#
